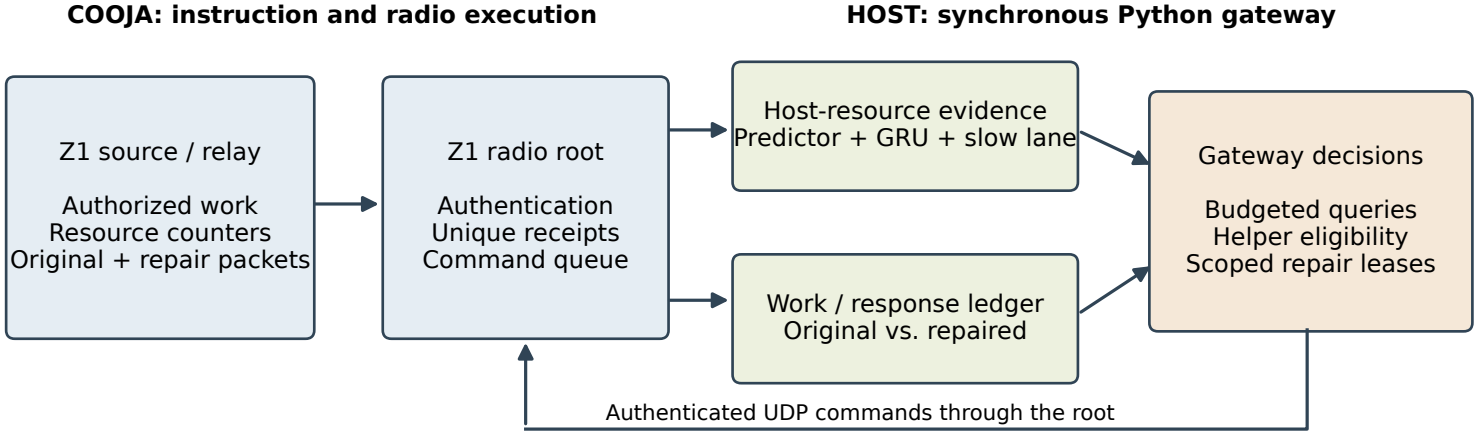
COOJA: instruction and radio execution
HOST: synchronous Python gateway
Z1 source / relay
Authorized work
Resource counters
Original + repair packets
Z1 radio root
Authentication
Unique receipts
Command queue
Host-resource evidence
Predictor + GRU + slow lane
Work / response ledger
Original vs. repaired
Gateway decisions
Budgeted queries
Helper eligibility
Scoped repair leases
Authenticated UDP commands through the root